CP01 SOFTWARE & APPLICATIONS

THE ULTIMATE FILE EXTENSION CHEAT SHEET

Kenali format fail, fahami kegunaannya dan tahu aplikasi yang sesuai untuk membukanya.

`.PDF` `.DOCX` `.CSV` `.XLSX` `.PNG` `.SVG` `.MP3` `.MP4` `.ZIP` `.JSON`

What Is a File Extension?

Suffix di hujung nama fail yang biasanya menunjukkan format atau jenis fail.



FORMAT
DOCX
Microsoft Word Open XML document format

FUNGSI
- Extension bantu operating system dan aplikasi kenal pasti jenis fail.
- Extension berbeza mungkin perlukan aplikasi berbeza.

INGAT
- Tukar nama extension **tidak** menukar format fail sebenar.
- Sesetengah format boleh dibuka oleh beberapa aplikasi.

TAK NAMPAK EXTENSION PADA NAMA FAIL?
- Windows 11: File Explorer, View, Show, tick `File name extensions` .
- macOS: Finder, Settings, Advanced, tick `Show all filename extensions` .

Quick Category Overview

Sembilan kategori utama dan contoh extension di dalamnya.

Documents HLM 2 `.PDF` `.DOCX` `.TXT`	**Spreadsheets** HLM 2 `.XLSX` `.CSV`	**Presentations** HLM 3 `.PPTX` `.ODP`
Images HLM 3 `.JPG` `.PNG` `.SVG`	**Audio** HLM 4 `.MP3` `.WAV` `.FLAC`	**Video** HLM 5 `.MP4` `.MOV` `.MKV`
Archives HLM 5 `.ZIP` `.RAR` `.7Z`	**Programming & Data** HLM 6 `.HTML` `.PY` `.JSON`	**Executables & Installers** HLM 7 `.EXE` `.APK` `.DMG`

CHEAT SHEET INI JAWAB LIMA SOALAN
- Apa maksud extension ini?
- Biasanya digunakan untuk apa?
- Aplikasi mana yang boleh membukanya?
- Apa bezanya dengan format yang serupa?
- Format mana patut dipilih untuk sesuatu tugas? Lihat halaman 8.

A Document Formats

Fail untuk teks, laporan dan dokumentasi.

EXTENSION	FULL NAME / MEANING	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.PDF	Portable Document Format	Sharing documents while preserving layout	Adobe Acrobat Reader, browser PDF viewers, Preview
.DOCX	Word Open XML Document	Editable word-processing documents	Microsoft Word, Google Docs, LibreOffice Writer
.DOC	Microsoft Word Binary Document	Older Word documents	Microsoft Word, LibreOffice Writer
.TXT	Plain Text File	Simple text without rich formatting	Notepad, TextEdit, VS Code
.RTF	Rich Text Format	Text with basic formatting	Word, WordPad where available, LibreOffice Writer
.ODT	OpenDocument Text	Open document word-processing format	LibreOffice Writer, OpenOffice Writer
.MD	Markdown	Structured plain-text writing and documentation	VS Code, Obsidian, Typora, Markdown editors

PDF VS DOCX VS TXT

PILIH IKUT TUJUAN

PDF
Best for sharing
- Preserve rupa dokumen.

DOCX
Best for editing
- Edit dokumen yang ada formatting.

TXT
Best for simple text
- Teks ringkas tanpa formatting.

PENTING
- PDF boleh ada elemen interaktif dan form, tetapi biasanya digunakan bila rupa dokumen perlu dikekalkan.

B Spreadsheet Formats

Fail untuk data berjadual, formula dan pengiraan.

EXTENSION	FULL NAME / MEANING	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.XLSX	Excel Open XML Spreadsheet	Spreadsheet with formulas, formatting and multiple sheets	Microsoft Excel, Google Sheets, LibreOffice Calc
.XLS	Older Excel Workbook	Legacy spreadsheet files	Microsoft Excel, LibreOffice Calc
.CSV	Comma-Separated Values	Tabular data exchange in plain text	Excel, Google Sheets, LibreOffice Calc, text editors
.ODS	OpenDocument Spreadsheet	Open spreadsheet format	LibreOffice Calc, OpenOffice Calc, compatible spreadsheet applications

CSV VS XLSX

DATA EXCHANGE VS FEATURE-RICH WORKBOOK

CSV
Plain-text tabular data
- Biasanya satu table dalam satu fail.
- Tidak simpan formula, formatting atau banyak worksheet.
- Delimiter boleh berbeza: comma, semicolon atau tab.

XLSX
Feature-rich spreadsheet
- Sokong banyak worksheet.
- Sokong formula dan formatting.
- Boleh ada chart, table dan feature spreadsheet lain.
- Guna Office Open XML package berasaskan XML.

PENTING
- CSV sesuai untuk pertukaran data. XLSX lebih sesuai untuk kerja dengan spreadsheet yang kaya feature.

C Presentation Formats

Fail untuk slide presentation.

EXTENSION	FULL NAME / MEANING	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.PPTX	PowerPoint Open XML Presentation	Modern slide presentations	Microsoft PowerPoint, Google Slides, LibreOffice Impress
.PPT	Older PowerPoint Presentation	Legacy presentations	Microsoft PowerPoint, LibreOffice Impress
.ODP	OpenDocument Presentation	Open presentation format	LibreOffice Impress, OpenOffice Impress

PPTX
Modern PowerPoint format

PPT
Older PowerPoint format

ODP
OpenDocument presentation format

IMAGE FILE FORMATS

Format imej berbeza dari segi compression, transparency, scalability, animation dan kualiti.

EXTENSION	FORMAT	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.JPG .JPEG	Joint Photographic Experts Group	Photographs and compressed images	Photos, Preview, Photoshop, browsers
.PNG	Portable Network Graphics	Images requiring lossless compression or transparency	Browsers, Photoshop, GIMP, Preview
.SVG	Scalable Vector Graphics	Logos, icons, diagrams and vector illustrations	Browsers, Figma, Illustrator, Inkscape
.GIF	Graphics Interchange Format	Simple animations and limited-colour graphics	Browsers, image viewers, editors
.WEBP	WebP	Web images with efficient compression	Modern browsers, compatible image editors
.HEIC	High Efficiency Image Container	Photos commonly captured by Apple devices	Apple Photos, Preview and compatible applications
.TIFF .TIF	Tagged Image File Format	High-quality image storage, scanning and professional workflows	Photoshop, Preview, GIMP, image viewers
.BMP	Bitmap Image File	Raster image storage, often uncompressed	Paint, Photoshop, GIMP

JPG vs PNG vs SVG

Tiga format yang paling kerap dibandingkan.

JPG
Suitable for photographs
- Uses lossy compression.
- Does not support transparency in the usual JPEG format.

PNG
Graphics and clean edges
- Uses lossless compression.
- Supports transparency.
- Suitable for graphics, screenshots and images requiring clean edges.

SVG
Logos, icons, diagrams
- Vector-based graphics format.
- Scales without the usual pixelation associated with raster images.
- Suitable for logos, icons and diagrams.

Raster vs Vector

Perbezaan cara imej dibina dan apa jadi bila dibesarkan.

RASTER

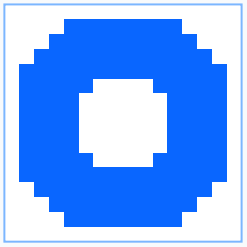
JPG, PNG, GIF, WebP: tersusun daripada pixels

SAIZ ASAL



ZOOM

DIBESARKAN



Tepi nampak bergerigi bila zoom

VECTOR

SVG: shapes dan paths yang dikira semula

SAIZ ASAL



ZOOM

DIBESARKAN



Tepi kekal licin pada mana-mana saiz

PENTING

- **Raster**: made of pixels. Examples: JPG, PNG, GIF, WebP.
- **Vector**: uses geometric shapes and paths. Example: SVG.
- Jangan anggap SVG sentiasa lebih baik daripada PNG atau JPG. Kesesuaian bergantung pada imej dan tujuan penggunaan.

A Audio Formats

Fail untuk muzik, rakaman dan audio.

EXTENSION	FORMAT	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.MP3	MPEG Audio Layer III	Compressed music and audio	VLC, Windows Media Player, music players
.WAV	Waveform Audio File Format	Audio recording and editing	Audacity, VLC, audio editors
.AAC	Advanced Audio Coding	Compressed audio	VLC, Apple Music and compatible players
.FLAC	Free Lossless Audio Codec	Lossless audio storage	VLC, foobar2000, compatible music players
.M4A	MPEG-4 Audio	Audio files, including AAC-encoded audio	Apple Music, VLC, compatible players
.OGG	Ogg Container	Audio and multimedia	VLC, compatible media players

MP3 VS WAV VS FLAC — LOSSY, UNCOMPRESSED, LOSSLESS

MP3

Everyday listening

- Lossy compression.
- Smaller file sizes.

WAV

Recording and editing

- Commonly stores uncompressed PCM audio.
- File sizes can be large.

FLAC

Lossless compression

- Preserves original audio data while reducing file size compared with uncompressed audio.

B Video Formats

Fail video ialah container yang memegang beberapa stream.

EXTENSION	FORMAT / CONTAINER	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.MP4	MPEG-4 Part 14 Container	General video sharing and playback	VLC, QuickTime, media players, browsers
.MOV	QuickTime Movie Container	Video production and Apple workflows	QuickTime, VLC, compatible editors
.MKV	Matroska Container	Video with multiple audio/subtitle tracks	VLC, compatible media players
.AVI	Audio Video Interleave	Older video container	VLC, compatible media players
.WEBM	WebM Container	Web-based video	Browsers, VLC
.WMV	Windows Media Video	Windows-oriented video workflows	Windows Media Player, VLC

VIDEO CONTAINER VS VIDEO CODEC — CONTOH: MP4 DENGAN H.264 ATAU H.265

MP4 CONTAINER

Container
Holds video, audio, subtitles and metadata.

Video stream
Encoded with a codec such as H.264 or H.265.

Codec
Encodes and decodes the video or audio streams.

PENTING

- MP4 dan MOV ialah container, bukan video codec. Keserasian playback bergantung pada codec dan feature yang digunakan dalam container.

A Archive & Compressed Files

Fail untuk package dan mampatkan banyak fail.

EXTENSION	FORMAT	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.ZIP	ZIP Archive	Compressing and packaging files	Windows File Explorer, macOS Archive Utility, 7-Zip
.RAR	RAR Archive	Compressed file archives	WinRAR, 7-Zip
.7Z	7-Zip Archive	High-compression file archives	7-Zip, compatible archive tools
.TAR	Tape Archive	Packaging files, commonly in Unix/Linux environments	tar, 7-Zip
.GZ	Gzip Compressed File	Compressing individual files or data streams	gzip, 7-Zip
.TAR.GZ .TGZ	Tar + Gzip	Packaged and compressed archives	tar, 7-Zip

Multiple files a.txt, b.png, c.pdf → **Archive** Dipackage jadi satu → **Compressed package** Saiz dikecilkan → **Extract** Unpack dengan tool → **Original files** Kembali seperti asal

BEZA TAR DAN GZ

- ZIP, RAR dan 7Z ialah archive formats.
- TAR package fail tetapi tidak menyediakan compression sendiri.
- GZ memampatkan data.
- TAR.GZ gabungkan packaging dan compression.

B Programming & Web Development Formats

Source code dan fail untuk web. Semua boleh dibuka dengan text editor.

EXTENSION	FORMAT / LANGUAGE	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.HTML	HyperText Markup Language	Webpage structure	Browser, VS Code
.CSS	Cascading Style Sheets	Webpage styling	VS Code, text editors
.JS	JavaScript	Web programming and scripting	VS Code, text editors
.TS	TypeScript	Typed JavaScript development	VS Code, compatible IDEs
.PY	Python	Python scripts	VS Code, PyCharm, text editors
.JAVA	Java Source File	Java source code	IntelliJ IDEA, Eclipse, VS Code
.C	C Source File	C programming	VS Code, Visual Studio, compatible IDEs
.CPP	C++ Source File	C++ programming	VS Code, Visual Studio, CLion
.PHP	PHP Source File	Server-side web development	VS Code, PhpStorm
.SQL	SQL Script	Database queries and commands	DBeaver, DataGrip, VS Code

C Data & Configuration Formats

Fail untuk data berstruktur, konfigurasi dan rekod.

EXTENSION	FORMAT	COMMON USE	APPLICATIONS THAT CAN OPEN IT
.JSON	JavaScript Object Notation	Data exchange and application configuration	VS Code, text editors, JSON tools
.XML	Extensible Markup Language	Structured data and document representation	VS Code, browsers, XML editors
.YAML .YML	YAML	Configuration and structured data	VS Code, text editors
.LOG	Log File	Application and system event records	Notepad, VS Code, log viewers
.DB	Database File (generic extension)	Database storage; actual format depends on the application	SQLite tools or application-specific software

PENTING

- JSON ialah data format, bukan programming language.
- HTML ialah markup language, bukan programming language.
- SQL ialah query language.
- `.DB` tidak mewakili satu format database universal. Database engine sebenar perlu diketahui.

Executables, Installers & Special Formats

Fail yang boleh pasang atau jalankan software.

EXTENSION	FORMAT	COMMON USE	APPLICATIONS / ENVIRONMENT
`.EXE`	Windows Executable	Running Windows applications	Windows
`.MSI`	Windows Installer Package	Installing or managing Windows software	Windows Installer
`.APK`	Android Package	Installing Android applications	Android
`.IPA`	iOS App Archive	Packaging iOS applications	Apple development/distribution ecosystem
`.DMG`	Apple Disk Image	Distributing macOS applications and files	macOS DiskImageMounter
`.ISO`	Disc Image	Representing optical disc contents	Windows, macOS, Linux image tools
`.DLL`	Dynamic-Link Library	Shared code and resources used by Windows applications	Windows applications and development tools
`.APP`	Application Bundle / Package	macOS application bundle or other platform-specific application packages	macOS, depending on context

Platform Quick Map

Extension mana untuk platform mana.

Windows
`.EXE` `.MSI` `.DLL`

macOS
`.DMG` `.APP`

Android
`.APK`

iOS
`.IPA`

Disc image
`.ISO`

PERHATIAN

BE CAREFUL WITH EXECUTABLE FILES

Files such as EXE, MSI and APK can install or execute software. Sebelum buka:

- Verify the source.
- Check the publisher or developer.
- Avoid unexpected email attachments.
- Do not bypass security warnings without understanding the risk.
- Download software from official sources.

Extension sahaja tidak menjamin fail itu selamat, atau fail itu benar-benar mengandungi format yang didakwa.

AWAS DOUBLE EXTENSION

`invoice.pdf` `.exe`

Extension sebenar ialah bahagian terakhir, iaitu `.exe`. Nama tu nampak macam PDF, tetapi ia fail executable.

SEMAK JENIS FAIL SEBENAR

Pada macOS dan Linux, command `file` menilai kandungan fail, bukan sekadar nama.

```
file report.pdf
```

A Formats That Are Easy to Confuse

Perbezaan utama antara pasangan format yang biasa dikelirukan.

COMPARISON	MAIN DIFFERENCE
PDF vs DOCX	Fixed-layout document sharing vs editable word-processing document
CSV vs XLSX	Plain-text tabular data vs feature-rich spreadsheet workbook
JPG vs PNG	Lossy photographic compression vs lossless graphics with transparency support
PNG vs SVG	Pixel-based raster image vs scalable vector graphic
MP3 vs WAV	Lossy compressed audio vs commonly uncompressed PCM audio
WAV vs FLAC	Commonly uncompressed audio vs losslessly compressed audio
MP4 vs MOV	Different multimedia containers with different compatibility/workflow patterns
ZIP vs RAR	Different archive formats with different implementations and compatibility
HTML vs CSS	Webpage structure vs presentation and styling
JSON vs XML	Different structured data representation formats

B Which Format Should You Choose?

Cadangan umum ikut situasi. Format terbaik bergantung pada aplikasi, audiens dan workflow.

SITUATION	RECOMMENDED FORMAT
Share a final document	`PDF`
Edit a written assignment	`DOCX`
Exchange simple tabular data	`CSV`
Work with formulas and multiple worksheets	`XLSX`
Create a logo for scalable use	`SVG`
Share a photograph	`JPG`
Share a transparent graphic	`PNG`
Store losslessly compressed audio	`FLAC`
Share a widely compatible video	`MP4` with compatible codecs
Package multiple files	`ZIP`
Share structured application data	`JSON`
Write a webpage	`HTML` `CSS` `JS`

IMPORTANT REMINDERS

1. Menukar `.docx` kepada `.pdf` dengan rename extension tidak menukar dokumen.
2. File extension ialah petunjuk yang berguna, tetapi bukan jaminan kandungan fail.
3. Satu format boleh disokong oleh beberapa aplikasi berbeza.
4. Sesetengah format perlukan software atau codec tertentu.
5. Sentiasa berhati-hati dengan executable files yang tidak dikenali.

Know the extension. Understand the format. Choose the right tool. TAKEAWAY

Save untuk rujukan bila jumpa format fail yang pelik.
Download PDF ini dan simpan sebagai rujukan pantas.

yusufaimxn
Technology, Made Understandable.

